

Haloalkane Substitution

Subject & topics: Chemistry | Organic | Reactions **Stage & difficulty:** A Level P1

Part A

2-Iodobutane and sodium ethoxide

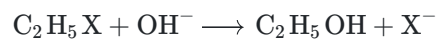
What is the product of a nucleophilic substitution reaction between 2-iodobutane and sodium ethoxide (NaOC_2H_5)?

- ☐ $\text{CH}_3\text{CH}=\text{CHCH}_3$
- ☐ $\text{CH}_3\text{CH}_2\text{CH}(\text{OCH}_3)\text{CH}_2\text{CH}_3$
- ☐ $\text{CH}_3\text{CH}_2\text{CH}(\text{CH}_3)\text{OCH}_2\text{CH}_3$
- ☐ $(\text{CH}_3)_2\text{CHCH}_2\text{OCH}_2\text{CH}_3$

Part B



Why does the reaction



take place more rapidly in aqueous solution when X is I than when X is Br?

- ☐ The I^- ion is a stronger nucleophile than the Br^- ion.
- ☐ The I^- ion is less hydrated in solution than the Br^- ion.
- ☐ The C–Br bond is weaker than the C–I bond.
- ☐ The C–Br bond is stronger than the C–I bond.

Part A adapted with permission from UCLES, A-Level Chemistry, June 1994, Paper 4, Question 22;

Part B adapted with permission from UCLES, A-Level Chemistry, June 1995, Paper 4, Question 20

Question deck:

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Haloalkane Substitution Mechanism

Subject & topics: Chemistry | Organic | Reactions **Stage & difficulty:** A Level P1

Part A

Energy profile

Halogenoalkanes react with aqueous alkali. One mechanism of this reaction has the energy profile shown below.

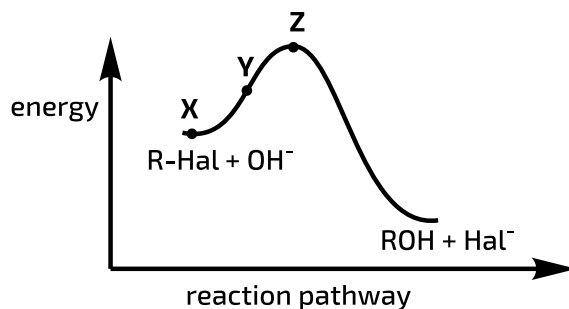


Figure 1: Energy profile for haloalkane with aqueous alkali

Which of the following statements are correct?

1. The reaction is an example of nucleophilic substitution.
2. Between X and Z the C–Hal bond will be lengthening.
3. The energy difference between X and Y represents the activation energy.

- ☐ 1, 2 and 3 are correct
- ☐ 1 and 2 only are correct
- ☐ 1 and 3 only are correct
- ☐ 2 and 3 only are correct
- ☐ 1 only is correct
- ☐ 2 only is correct
- ☐ 3 only is correct

Part B

Curly arrow mechanism

Which diagram correctly represents the transfer of electrons when ammonia reacts with a chloroalkane (alkyl chloride)?

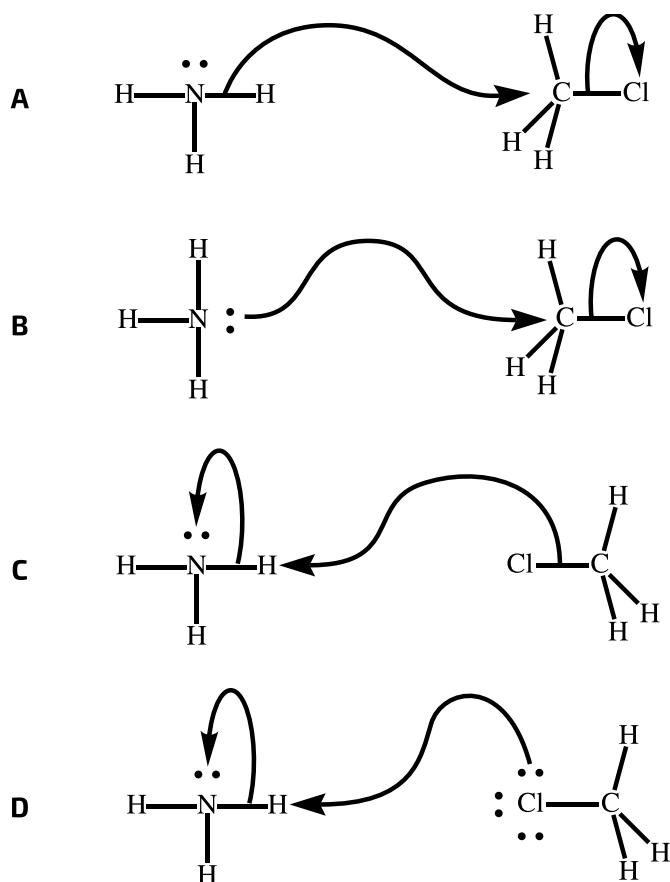


Figure 2: Possible mechanisms for ammonia with chloroalkane

- ☐ A
- ☐ B
- ☐ C
- ☐ D

Part A adapted with permission from UCLES, A-Level Chemistry, June 1993, Paper 4, Question 37;

Part B adapted with permission from UCLES, A-Level Chemistry, June 1994, Paper 4, Question 26

Alkene Bromination

Subject & topics: Chemistry | Organic | Reactions

Stage & difficulty: A Level P1

Part A

Bromination of C_4H_8

Which of the following compounds could be formed by the action of bromine on an alkene of formula C_4H_8 ?

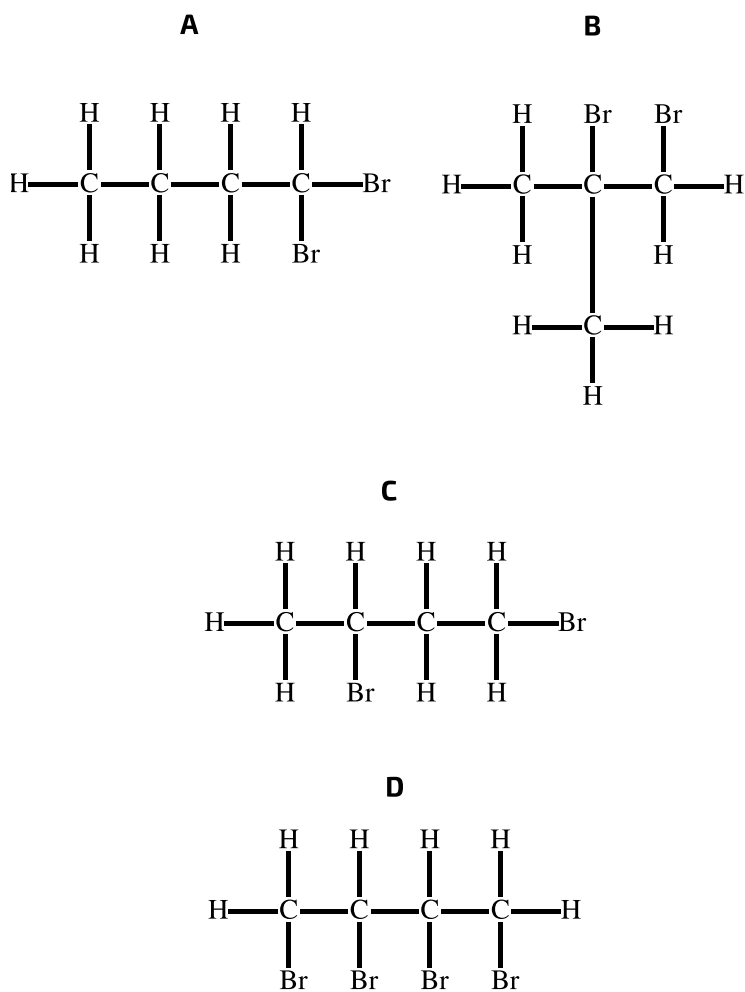


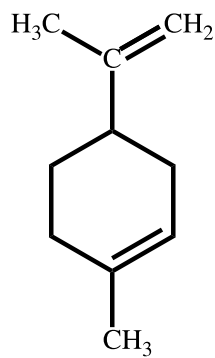
Figure 1: Possible products of C_4H_8 with bromine

- ☐ A
- ☐ B
- ☐ C
- ☐ D

Part B

Bromination of limonene

Limonene is an oil formed in the peel of citrus fruits.



Limonene

Figure 2: Structure of limonene

Which product is formed when limonene reacts with excess molecular bromine at room temperature in the dark?

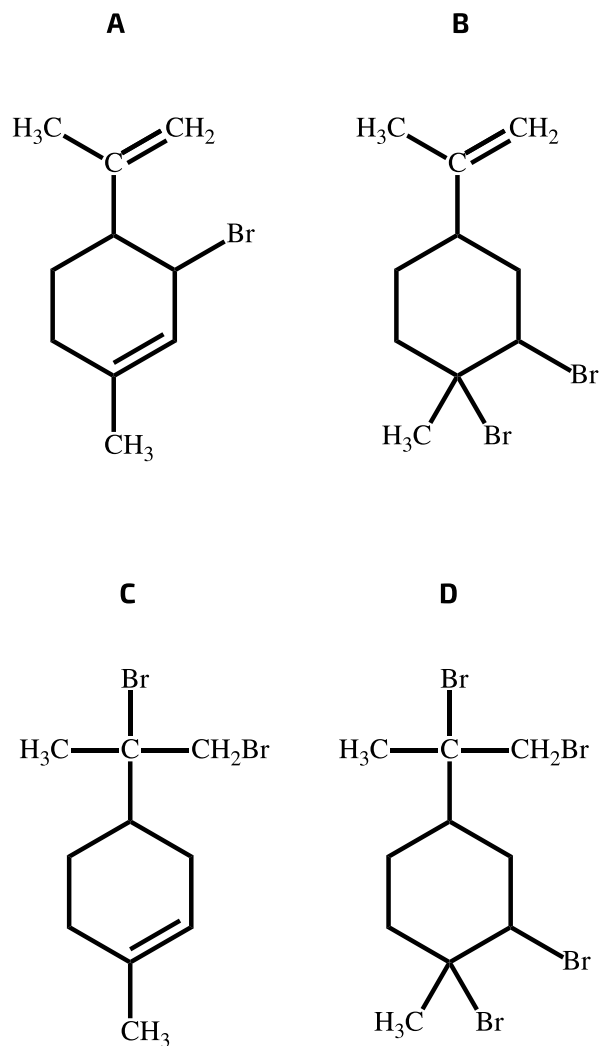


Figure 3: Possible products of limonene bromination

- ☐ A
- ☐ B
- ☐ C
- ☐ D

Part A adapted with permission from UCLES, A-Level Chemistry, November 1994, Paper 4, Question 22;
 Part B adapted with permission from UCLES, A-Level Chemistry, November 1996, Paper 4, Question 22

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Alkene Bromination Mechanism

Subject & topics: Chemistry | Organic | Reactions Stage & difficulty: A Level P1

Part A

Curly arrows

What is the sequence of curly arrows denoting movement of electrons in the first step of the reaction between ethene and bromine (below)?

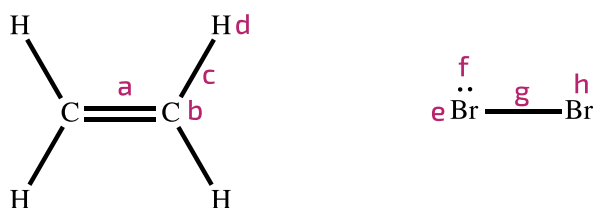


Figure 1: Bromination mechanism

For example in the reaction below, if you think the mechanism is as shown, your answer would be **cdef**.

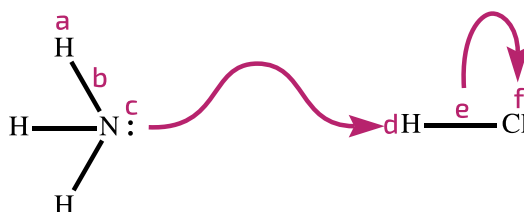


Figure 2: Sequence **cdef**

Ethene bromination sequence:

Part B

Reaction type

What type of reaction is this?

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Alkenes With HBr

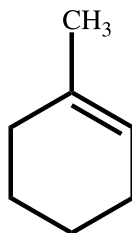
Subject & topics: Chemistry | Organic | Reactions

Stage & difficulty: A Level P2

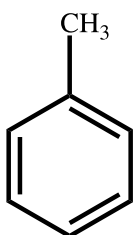
Part A

1-Methylcyclohexene with HBr

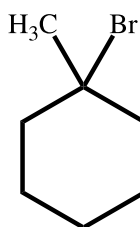
Which of the following structures represents the main organic compound produced when hydrogen bromide is added to 1-methylcyclohexene (shown below)?



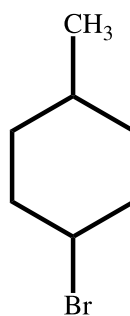
A



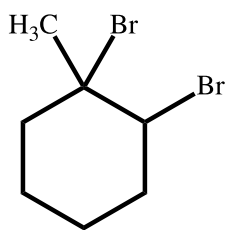
B



C



D



E

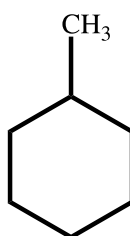


Figure 1: 1-methylcyclohexene and possible products of reaction with HBr

- ☐ A
- ☐ B
- ☐ C
- ☐ D
- ☐ E

Part B

Methylpropene with HBr

Methylpropene has the structure shown below.

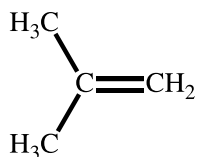


Figure 2: Structure of methylpropene

What is the **major** product formed when methylpropene reacts with HBr?

Use the [structure editor](#) to generate a SMILES string.

In the editor, after drawing your structure, click on the round, yellow smiley face to generate a SMILES string. Copy the SMILES string and paste it in the answer box.

[Using the structure editor](#)

Part A adapted with permission from OCSEB, A-Level Chemistry, June 1998, Paper 1, Question 22;
Part B created for Isaac Physics by R. Less

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Alkene Reactions

Subject & topics: Chemistry | Organic | Reactions **Stage & difficulty:** A Level P2

Propene reacts under the following conditions to give compounds **A**, **B** and **C**.

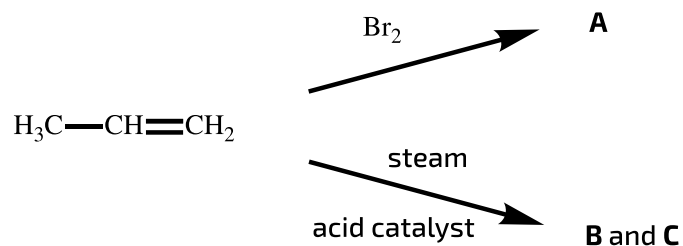


Figure 1: Propene with bromine and steam in presence of an acid catalyst

Part A with Br₂

What is product **A**?

Use the [structure editor](#) to generate a SMILES string.

In the editor, after drawing your structure, click on the round, yellow smiley face to generate a SMILES string. Copy the SMILES string and paste it in the answer box.

[Using the structure editor](#)

Part B

with steam / acid catalyst

What are products **B** and **C**?

Use the [structure editor](#) to generate SMILES strings.

Give your answer in the format "**B, C**" (space after comma).

In the editor, after drawing your structure, click on the round, yellow smiley face to generate a SMILES string. Copy the SMILES string and paste it in the answer box.

[Using the structure editor](#)

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More Alkene Bromination

Subject & topics: Chemistry | Organic | Reactions **Stage & difficulty:** A Level P1

Compound **A** undergoes the following reactions:



Part A

Bromination of compound A

What is compound **A**? (There are two possible isomers: give the structure of either.)

Use the [structure editor](#) to generate a SMILES string.

Click on the round, yellow smiley face to generate a SMILES string. Copy the SMILES string and paste it in the answer box.

[Using the structure editor](#)

Part B

Step (II)

What type of reaction is step (II)?

Part C

Ethene with aqueous bromine

Ethene reacts with aqueous bromine to give the two products $\text{CH}_2\text{BrCH}_2\text{Br}$ and $\text{CH}_2\text{BrCH}_2\text{OH}$.

Which statement is correct for these products?

- ☐ Reaction of ethene with aqueous HBr gives the same products.
- ☐ Both products are obtained in this reaction by nucleophilic addition.
- ☐ Both products possess an overall dipole.
- ☐ Both products can be hydrolysed to form the same diol.
- ☐ Both products are obtained in this reaction by electrophilic substitution.

Part A adapted with permission from UCLES, A-Level Chemistry, June 1991, Paper 3, Question 8;

Part B adapted with permission from UCLES, A-Level Chemistry, November 1993, Paper 4, Question 24

Question deck:

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Free Radical Reactions

Subject & topics: Chemistry | Organic | Reactions **Stage & difficulty:** A Level P1

Part A

CFCs

In the upper atmosphere, chlorofluoroalkanes (CFCs) are broken down to give chlorine radicals but not fluorine radicals.

What is the best explanation for this?

- ☐ Fluorine has a lower atomic number.
- ☐ Chlorine has a higher molecular weight.
- ☐ The C–F bond is stronger than the C–Cl bond.
- ☐ Fluorine has a higher ionisation energy than chlorine.

Part B

Chlorination of methane

Methane reacts with chlorine in the presence of sunlight. Which statement about the intermediates is correct?

- ☐ They are more energetically stable than the reactants.
- ☐ They are positively charged ions.
- ☐ They combine to form HCl.
- ☐ They contain an odd number of electrons.

Question deck:

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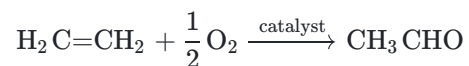
Ethene Oxidation

Subject & topics: Chemistry | Organic | Reactions **Stage & difficulty:** A Level C1

Part A

Ethene to ethanal

Aldehydes and ketones are produced industrially by the catalytic oxidation of alkenes, e.g. ethanal is manufactured from ethene as shown below:



This process is also used industrially with but-2-ene.

Which of the following represents the structure of the compound which would be produced from but-2-ene?

- ☐ $\text{CH}_3\text{CH}_2\text{CHO}$
- ☐ $\text{CH}_3\text{CH}_2\text{CH}_2\text{CHO}$
- ☐ $\text{CH}_3\text{COCH}_2\text{CH}_3$
- ☐ CH_3COCH_3
- ☐ $(\text{CH}_3)_2\text{CHCHO}$

Ethene with bromine and sodium nitrate

When ethene reacts with bromine in the presence of concentrated aqueous sodium nitrate, the product contains the following compound:

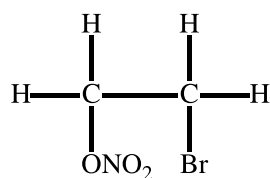


Figure 1: Reaction product

What is the intermediate formed in this reaction?

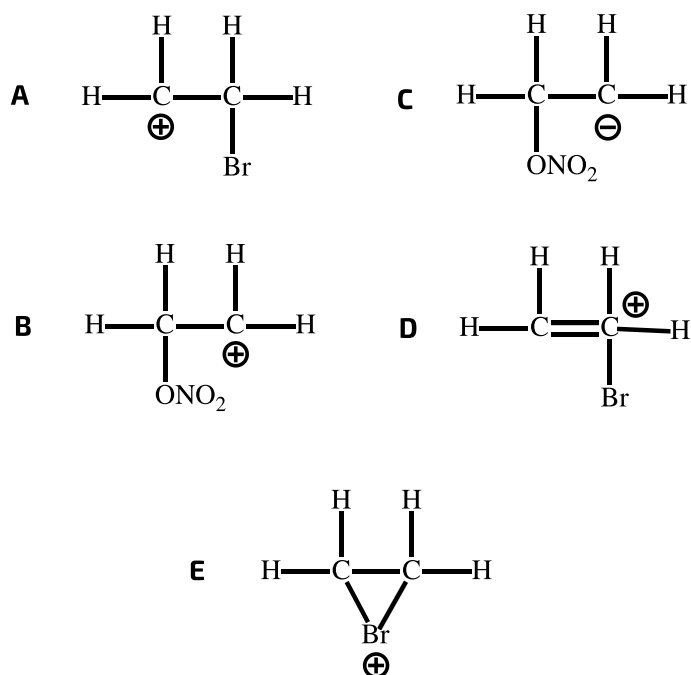


Figure 2: Possible intermediates

- ☐ A
- ☐ B
- ☐ C
- ☐ D
- ☐ E

Part A adapted with permission from UCLES, A-Level Chemistry, November 1993, Paper 4, Question 22;
Part B adapted with permission from UCLES, A-Level Chemistry, November 1995, Paper 4, Question 22

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Epoxy Precursor

Subject & topics: Chemistry | Organic | Reactions **Stage & difficulty:** A Level C1

Epoxy resins are polymers which are used as adhesives. One monomer used in their manufacture has the displayed formula:

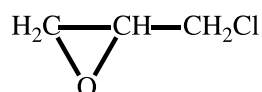


Figure 1: Epoxy resin monomer

This is manufactured from propene in three stages:

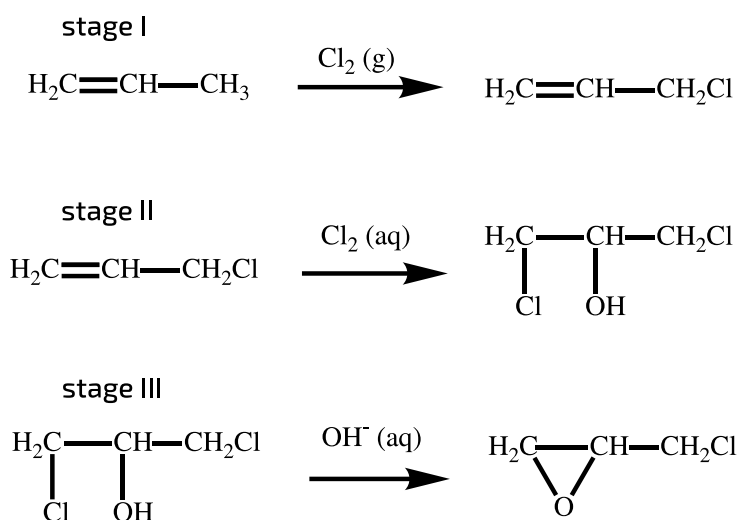


Figure 2: Three stages of epoxy monomer manufacture from propene

Part A
Stage I

What type of reaction mechanism takes place between propene and chlorine gas in stage I?

Suggest what conditions are necessary for this reaction.

Part B
Stage II

What type of reaction mechanism takes place when the aqueous chlorine reacts in stage II?

Part C
Stage III

What type of organic reaction takes place in stage III?

Adapted with permission from UCLES, A-Level Chemistry, November 1994, Paper 3, Question 4

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